Evaluation of Vector Search Methods with FAISS

0. Environment and configuration

In [2]:
```
import os, time, json, gc
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import faiss

print("FAISS version :", faiss.__version__)
print("NumPy version :", np.__version__)

RNG_SEED = 1234
np.random.seed(RNG_SEED)
faiss.omp_set_num_threads(min(8, os.cpu_count() or 1))
print("FAISS threads :", faiss.omp_get_max_threads())
```

```
FAISS version : 1.14.2
NumPy version : 2.4.6
FAISS threads : 8
```

In [3]:
```
DATA_ROOT = "./data"
PROFILE   = "full"

RESULTS_DIR, PLOTS_DIR, QUERY_DIR = "./results", "./plots", "./modified_queries"
for _d in (RESULTS_DIR, PLOTS_DIR, QUERY_DIR):
    os.makedirs(_d, exist_ok=True)

K = 10

PROFILES = {
    "smoke": dict(max_base=50_000,  max_query=200,  nlist=1024),
    "full":  dict(max_base=None,    max_query=1000, nlist=4096),
}
CFG = PROFILES[PROFILE]
print(f"Profile = {PROFILE}  ->  {CFG}")
```

```
Profile = full  ->  {'max_base': None, 'max_query': 1000, 'nlist': 4096}
```

1. Loading the datasets

In [4]:
```
def read_fvecs(path, max_rows=None):
    a = np.fromfile(path, dtype="int32")
    if a.size == 0:
        raise ValueError(f"empty file: {path}")
    d = a[0]
    a = a.reshape(-1, d + 1)
    if max_rows is not None:
        a = a[:max_rows]
    return np.ascontiguousarray(a[:, 1:]).view("float32")
```

```python
def read_bvecs(path, max_rows=None):
    a = np.fromfile(path, dtype="uint8")
    d = a[:4].view("int32")[0]
    a = a.reshape(-1, d + 4)
    if max_rows is not None:
        a = a[:max_rows]
    return np.ascontiguousarray(a[:, 4:]).astype("float32")

def read_ivecs(path, max_rows=None):
    a = np.fromfile(path, dtype="int32")
    d = a[0]
    a = a.reshape(-1, d + 1)
    if max_rows is not None:
        a = a[:max_rows]
    return np.ascontiguousarray(a[:, 1:])
```

In [5]:
```python
def l2_normalize(x):
    x = np.ascontiguousarray(x.astype("float32", copy=True))
    faiss.normalize_L2(x)
    return x


def load_sift():
    base = read_fvecs(os.path.join(DATA_ROOT, "sift", "sift_base.fvecs"),
                      CFG["max_base"])
    query = read_fvecs(os.path.join(DATA_ROOT, "sift", "sift_query.fvecs"),
                       CFG["max_query"])
    return base, query

def load_gist():
    base = read_fvecs(os.path.join(DATA_ROOT, "gist", "gist_base.fvecs"),
                      CFG["max_base"])
    query = read_fvecs(os.path.join(DATA_ROOT, "gist", "gist_query.fvecs"),
                       CFG["max_query"])
    return base, query

def load_glove():
    """GloVe: expects glove_base.fvecs / glove_query.fvecs (see README).
    Vectors are L2-normalised so inner product == cosine similarity."""
    base = read_fvecs(os.path.join(DATA_ROOT, "glove", "glove_base.fvecs"),
                      CFG["max_base"])
    query = read_fvecs(os.path.join(DATA_ROOT, "glove", "glove_query.fvecs"),
                       CFG["max_query"])
    return l2_normalize(base), l2_normalize(query)

DATASETS = {
    "SIFT":  dict(loader=load_sift,  metric=faiss.METRIC_L2),
    "GIST":  dict(loader=load_gist,  metric=faiss.METRIC_L2),
    "GloVe": dict(loader=load_glove, metric=faiss.METRIC_INNER_PRODUCT),
}
```

In [6]:
```python
LOADED = {}
for name, spec in DATASETS.items():
    try:
        xb, xq = spec["loader"]()
        LOADED[name] = dict(xb=xb, xq=xq, metric=spec["metric"])
        print(f"{name:6s}: base={xb.shape}  query={xq.shape}  "
              f"metric={'IP' if spec['metric']==faiss.METRIC_INNER_PRODUCT else
    except Exception as e:
```

```
        print(f"{name:6s}: SKIPPED ({type(e).name}: {e})")

REPORT_ONLY = (len(LOADED) == 0)
if REPORT_ONLY:
    print("\nNo datasets found locally — running in REPORT-ONLY mode "
          "(plots and tables will still render from CSVs).")
    _csv = os.path.join(RESULTS_DIR, "sweep_results.csv")
    if not os.path.exists(_csv):
        raise RuntimeError("Neither datasets nor results/sweep_results.csv "
                           "are available — nothing to report.")
    _names = list(pd.read_csv(_csv).dataset.unique())
    LOADED = {n: dict(xb=None, xq=None,
                      metric=DATASETS[n]["metric"]) for n in _names}
```

```
SIFT  : base=(1000000, 128)  query=(1000, 128)  metric=L2
GIST  : base=(1000000, 960)  query=(1000, 960)  metric=L2
GloVe : base=(1183514, 100)  query=(1000, 100)  metric=IP
```

2. Ground truth and evaluation metrics

Ground truth. For each query we need the *exact* `k` nearest neighbours. We obtain them with a brute-force `IndexFlat` index (`IndexFlatL2` for Euclidean datasets, `IndexFlatIP` for normalised GloVe). This is the reference set $G_k(q)$ against which approximate results are scored.

Recall@k. If $A_k(q)$ is the approximate result set and $G_k(q)$ the exact set, then $\text{Recall@}k(q) = |A_k(q) \cap G_k(q)|/k$, averaged over all queries.

QPS. Throughput is $\text{QPS} = N/T$ where N is the number of queries and T the total wall-clock search time. We run the search a few times and keep the **best** time, which is the standard ANN-Benchmarks convention and reduces noise from background processes.

In [7]:
```
def build_ground_truth(xb, xq, k, metric):
    if metric == faiss.METRIC_INNER_PRODUCT:
        gt = faiss.IndexFlatIP(xb.shape[1])
    else:
        gt = faiss.IndexFlatL2(xb.shape[1])
    gt.add(xb)
    D_gt, I_gt = gt.search(xq, k)
    return D_gt, I_gt

def recall_at_k(I_approx, I_gt, k):
    hits = 0
    for i in range(I_gt.shape[0]):
        hits += len(set(I_approx[i, :k]).intersection(set(I_gt[i, :k])))
    return hits / (I_gt.shape[0] * k)

def per_query_recall(I_approx, I_gt, k):
    out = np.empty(I_gt.shape[0], dtype="float64")
    for i in range(I_gt.shape[0]):
        out[i] = len(set(I_approx[i, :k]).intersection(set(I_gt[i, :k]))) / k
    return out

def benchmark_index(index, xq, k, repeats=3):
    best_t, D, I = float("inf"), None, None
```

```
    for _ in range(repeats):
        t0 = time.perf_counter()
        D_, I_ = index.search(xq, k)
        t1 = time.perf_counter()
        if (t1 - t0) < best_t:
            best_t, D, I = (t1 - t0), D_, I_
    return D, I, best_t, xq.shape[0] / best_t
```

3. The three index families and what we sweep

Each method exposes a **structural** parameter (fixed when the index is built) and a **search-time** parameter (changed per query batch without rebuilding). The brief asks us to keep these two effects clearly separated, so the sweeps below do exactly that.

3.1 IVFPQ — inverted file + product quantization

`IVF{nlist},PQ{m}x8` partitions the space into `nlist` Voronoi cells and stores each vector as `m` product-quantization codes of 8 bits each.

- **`nprobe`** (search-time) — how many inverted lists are visited per query. Larger `nprobe` examines more of the database, raising recall and query time.
- **`m`** (structural) — the number of PQ sub-spaces. More sub-spaces means finer quantization (lower approximation error) but more memory. `m` must divide the dimension `d`; we pick the valid values among 8, 16, 32.

3.2 HNSWSQ — graph + scalar quantization

`HNSW32_SQ{b}` builds a 32-neighbour hierarchical navigable small-world graph over scalar-quantized vectors.

- **`efSearch`** (search-time) — size of the candidate list during graph traversal. Larger `efSearch` explores more of the graph, raising recall and query time.
- **`efConstruction`** (structural) — candidate list size while building the graph; a larger value yields a better-connected graph at higher build cost.
- **`b`** (structural) — scalar-quantization bit-width. `SQ4` / `SQ6` / `SQ8` trade memory against approximation error.

3.3 LSH — locality-sensitive hashing

`IndexLSH(d, nbits)` projects each vector onto `nbits` random hyperplanes and stores the resulting binary code; search ranks by Hamming distance.

- **`nbits`** (structural) — the number of hash bits / projections. More bits make codes more selective (better discrimination) but lower the collision probability, which can hurt recall. LSH has no search-time knob.

In [8]:
```
def valid_pq_m(d, targets=(8, 16, 32)):
    divisors = [m for m in range(2, d + 1) if d % m == 0]
    chosen = []
    for t in targets:
```

```
        m = min(divisors, key=lambda x: (abs(x - t), x))
        if m not in chosen:
            chosen.append(m)
    return sorted(chosen)

NPROBE_VALUES    = [1, 4, 8, 16, 32, 64]
EFSEARCH_VALUES = [16, 32, 64, 128, 256]
SQ_BITS         = [4, 6, 8]
LSH_NBITS       = [128, 256, 512]
EF_CONSTRUCTION = 200
```

In [9]:

```
def build_ivfpq(xb, m, metric, nlist):
    d = xb.shape[1]
    index = faiss.index_factory(d, f"IVF{nlist},PQ{m}x8", metric)
    if not index.is_trained:
        index.train(xb)
    index.add(xb)
    return index

def build_hnswsq(xb, b, metric, ef_construction=EF_CONSTRUCTION):
    d = xb.shape[1]
    index = faiss.index_factory(d, f"HNSW32_SQ{b}", metric)
    try:
        index.hnsw.efConstruction = ef_construction
    except AttributeError:
        pass
    if not index.is_trained:
        index.train(xb)
    index.add(xb)
    return index

def build_lsh(xb, nbits, metric):
    d = xb.shape[1]
    index = faiss.IndexLSH(d, nbits)
    index.train(xb)
    index.add(xb)
    return index

def set_search_param(index, name, value):
    faiss.ParameterSpace().set_index_parameter(index, name, value)
```

4. Main experiment — recall vs. QPS sweeps

For every dataset we:

1. build the exact ground truth once;
2. for **IVFPQ**, build one index per valid `m` and sweep `nprobe`;
3. for **HNSWSQ**, build one index per `SQ` bit-width and sweep `efSearch`;
4. for **LSH**, build one index per `nbits` (single search point each).

Every row of the results table records the method, dataset, the swept parameters, Recall@10 and QPS. The table is saved to `results/sweep_results.csv`.

In [10]:

```
def run_all_sweeps(loaded, k=K):
    rows = []
    for name, data in loaded.items():
```

```
    xb, xq, metric = data["xb"], data["xq"], data["metric"]
    d = xb.shape[1]
    print(f"\n=== {name}  (d={d}, base={xb.shape[0]}, query={xq.shape[0]}) =

    t0 = time.perf_counter()
    _, I_gt = build_ground_truth(xb, xq, k, metric)
    print(f"  ground truth built in {time.perf_counter()-t0:.1f}s")

    for m in valid_pq_m(d):
        t0 = time.perf_counter()
        idx = build_ivfpq(xb, m, metric, CFG["nlist"])
        build_t = time.perf_counter() - t0
        for nprobe in NPROBE_VALUES:
            set_search_param(idx, "nprobe", nprobe)
            D, I, st, qps = benchmark_index(idx, xq, k)
            rows.append(dict(method="IVFPQ", dataset=name, d=d,
                             param1=f"m={m}", param2=f"nprobe={nprobe}",
                             m=m, nprobe=nprobe, sq_bits=None, nbits=None,
                             recall=recall_at_k(I, I_gt, k), qps=qps,
                             build_time=build_t))
        print(f"  IVFPQ m={m:2d}  built in {build_t:.1f}s, "
              f"nprobe sweep done")
        del idx; gc.collect()

    for b in SQ_BITS:
        t0 = time.perf_counter()
        idx = build_hnswsq(xb, b, metric)
        build_t = time.perf_counter() - t0
        for ef in EFSEARCH_VALUES:
            set_search_param(idx, "efSearch", ef)
            D, I, st, qps = benchmark_index(idx, xq, k)
            rows.append(dict(method="HNSWSQ", dataset=name, d=d,
                             param1=f"SQ{b}", param2=f"efSearch={ef}",
                             m=None, nprobe=None, sq_bits=b, nbits=None,
                             recall=recall_at_k(I, I_gt, k), qps=qps,
                             build_time=build_t))
        print(f"  HNSWSQ SQ{b}  built in {build_t:.1f}s, "
              f"efSearch sweep done")
        del idx; gc.collect()

    for nbits in LSH_NBITS:
        t0 = time.perf_counter()
        idx = build_lsh(xb, nbits, metric)
        build_t = time.perf_counter() - t0
        D, I, st, qps = benchmark_index(idx, xq, k)
        rows.append(dict(method="LSH", dataset=name, d=d,
                         param1=f"nbits={nbits}", param2="-",
                         m=None, nprobe=None, sq_bits=None, nbits=nbits,
                         recall=recall_at_k(I, I_gt, k), qps=qps,
                         build_time=build_t))
        print(f"  LSH nbits={nbits:4d}  built in {build_t:.1f}s")
        del idx; gc.collect()

return pd.DataFrame(rows)
```

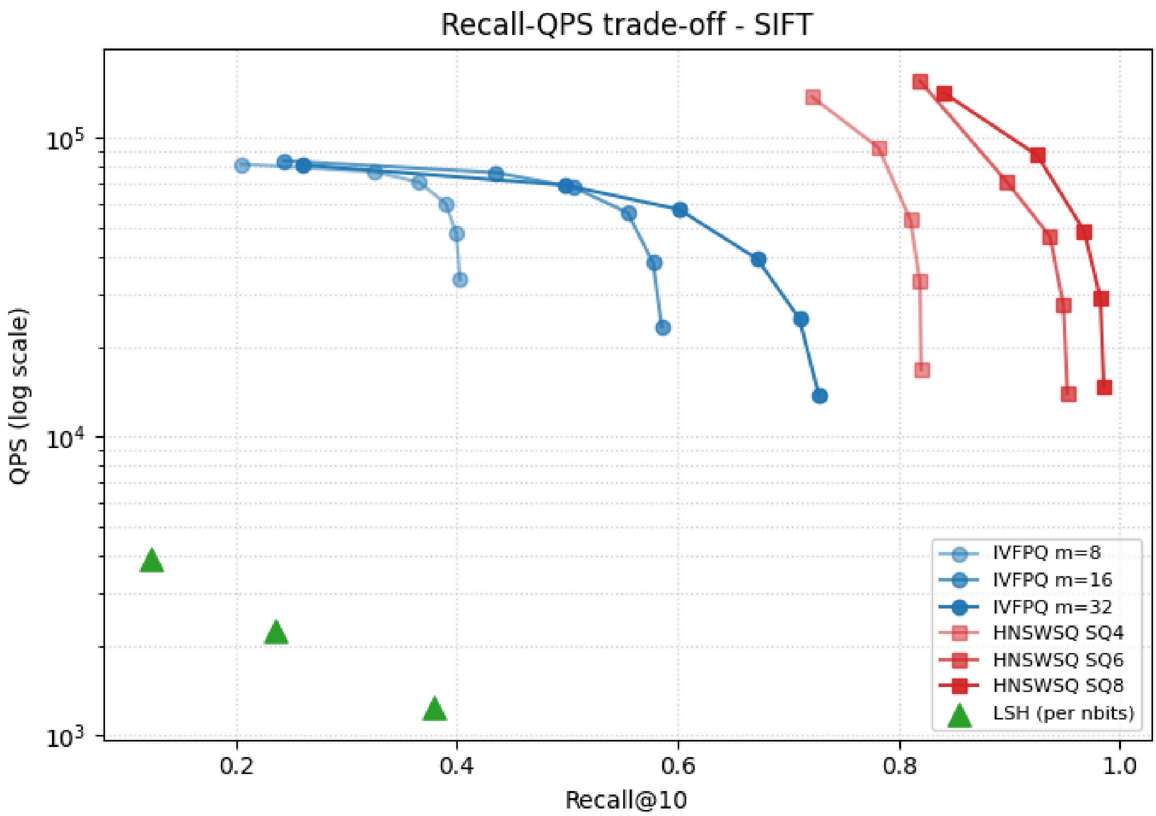
5. The recall–QPS trade-off

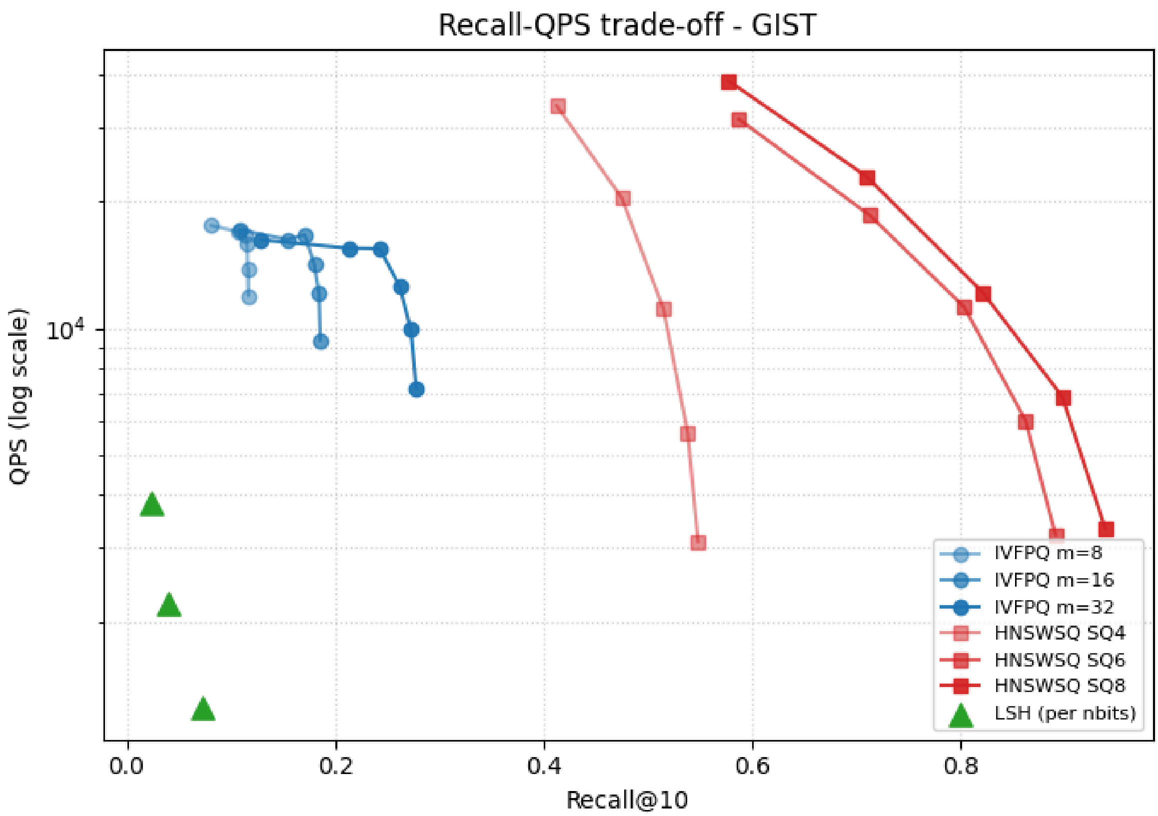
A good ANN method is **not** simply the one with the highest recall or the highest QPS, it is the one that sits on the best **trade-off frontier**. The plots below put Recall@10 on the x-axis and QPS on the y-axis. Points towards the **top-right** are better. A method whose curve lies above and to the right of another *dominates* it.

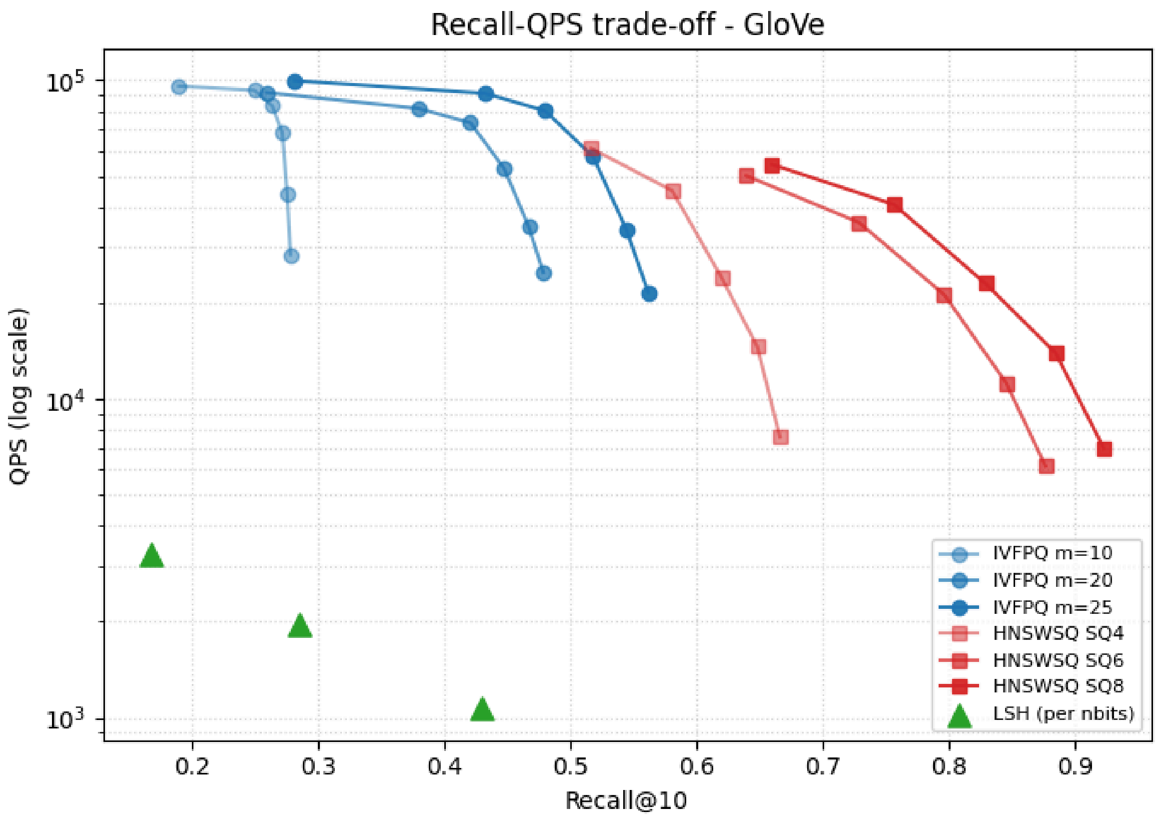
Each `IVFPQ` curve corresponds to one `m` traced over `nprobe` ; each `HNSWSQ` curve corresponds to one `SQ` width traced over `efSearch` ; `LSH` contributes one point per `nbits` .

In [11]:
```
MARKERS = {"IVFPQ": "o", "HNSWSQ": "s", "LSH": "^"}
COLORS  = {"IVFPQ": "#1f77b4", "HNSWSQ": "#d62728", "LSH": "#2ca02c"}

def _shade(values, v):
    vals = sorted(set(values))
    if len(vals) <= 1:
        return 0.85
    return 0.5 + 0.45 * vals.index(v) / (len(vals) - 1)

def plot_tradeoff(df, dataset, ax=None):
    own_fig = ax is None
    if own_fig:
        fig, ax = plt.subplots(figsize=(7, 5))
    sub = df[df.dataset == dataset]

    ivf_ms = sorted(int(x) for x in sub.m.dropna().unique())
    for m in ivf_ms:
        d = sub[(sub.method == "IVFPQ") & (sub.m == m)].sort_values("recall")
        ax.plot(d.recall, d.qps, marker="o", color=COLORS["IVFPQ"],
                alpha=_shade(ivf_ms, m), label=f"IVFPQ m={m}")
    sq_bs = sorted(int(x) for x in sub.sq_bits.dropna().unique())
    for b in sq_bs:
        d = sub[(sub.method == "HNSWSQ") & (sub.sq_bits == b)].sort_values("reca
        ax.plot(d.recall, d.qps, marker="s", color=COLORS["HNSWSQ"],
                alpha=_shade(sq_bs, b), label=f"HNSWSQ SQ{b}")
    d = sub[sub.method == "LSH"].sort_values("recall")
    ax.scatter(d.recall, d.qps, marker="^", color=COLORS["LSH"],
               s=90, zorder=5, label="LSH (per nbits)")

    ax.set_yscale("log")
    ax.set_xlabel("Recall@10")
    ax.set_ylabel("QPS (log scale)")
    ax.set_title(f"Recall-QPS trade-off - {dataset}")
    ax.grid(True, which="both", ls=":", alpha=0.5)
    ax.legend(fontsize=8, loc="best")
    if own_fig:
        fig.tight_layout()
        fig.savefig(os.path.join(PLOTS_DIR, f"tradeoff_{dataset}.png"), dpi=150)
        plt.show()
```

In [12]:
```
results = pd.read_csv(os.path.join(RESULTS_DIR, "sweep_results.csv"))

for ds in results.dataset.unique():
    plot_tradeoff(results, ds)
```


Recall-QPS trade-off - SIFT
QPS (log scale)
10^5
10^4
10^3
0.2
0.4
0.6
0.8
1.0
Recall@10
IVFPQ m=8
IVFPQ m=16
IVFPQ m=32
HNSWSQ SQ4
HNSWSQ SQ6
HNSWSQ SQ8
LSH (per nbits)


Recall-QPS trade-off - GIST
QPS (log scale)
10^4
0.0
0.2
0.4
0.6
0.8
Recall@10
IVFPQ m=8
IVFPQ m=16
IVFPQ m=32
HNSWSQ SQ4
HNSWSQ SQ6
HNSWSQ SQ8
LSH (per nbits)



5.1 Best configuration per dataset

For a fair single-number comparison we report, for each method, the configuration with the **highest QPS that still reaches Recall@10 ≥ 0.90**. If no configuration reaches 0.90 we fall back to the highest-recall configuration and flag it.

In [13]:
```
def best_configs(df, recall_target=0.90):
    out = []
    for (ds, method), g in df.groupby(["dataset", "method"]):
        ok = g[g.recall >= recall_target]
        if len(ok):
            row = ok.loc[ok.qps.idxmax()].copy(); row["met_target"] = True
        else:
            row = g.loc[g.recall.idxmax()].copy(); row["met_target"] = False
        out.append(row)
    cols = ["dataset", "method", "param1", "param2", "recall", "qps",
            "met_target"]
    return pd.DataFrame(out)[cols].sort_values(["dataset", "method"])

best = best_configs(results)
best.to_csv(os.path.join(RESULTS_DIR, "best_configs.csv"), index=False)
best
```

Out[13]:

	dataset	method	param1	param2	recall	qps	met_target
83	GIST	HNSWSQ	SQ8	efSearch=256	0.9388	3322.880127	True
35	GIST	IVFPQ	m=32	nprobe=64	0.2778	7186.499470	False
104	GIST	LSH	nbits=512	-	0.0722	1243.841426	False
98	GloVe	HNSWSQ	SQ8	efSearch=256	0.9226	6994.501615	True
53	GloVe	IVFPQ	m=25	nprobe=64	0.5615	21613.287063	False
107	GloVe	LSH	nbits=512	-	0.4296	1066.170398	False
65	SIFT	HNSWSQ	SQ8	efSearch=32	0.9255	87299.685251	True
17	SIFT	IVFPQ	m=32	nprobe=64	0.7276	13782.406213	False
101	SIFT	LSH	nbits=512	-	0.3790	1233.633081	False

5.2 Cross-dataset winner — peak recall and runner-up flip

In [14]:

```
peak = (results
        .loc[results.groupby(["dataset", "method"])["recall"].idxmax()]
        [["dataset", "method", "param1", "param2", "recall", "qps"]]
        .reset_index(drop=True))

peak["rank"]   = peak.groupby("dataset")["recall"].rank(method="dense",
                                                       ascending=False).astype(
peak["winner"] = peak["rank"].map({1: "1st", 2: "2nd", 3: "3rd"})

peak = peak.sort_values(["dataset", "rank"]).reset_index(drop=True)

peak[["dataset", "winner", "method", "param1", "param2", "recall", "qps"]] \
    .to_csv(os.path.join(RESULTS_DIR, "cross_dataset_winners.csv"), index=False)

peak[["dataset", "winner", "method", "param1", "param2", "recall", "qps"]]
```

Out[14]:

	dataset	winner	method	param1	param2	recall	qps
0	GIST	1st	HNSWSQ	SQ8	efSearch=256	0.9388	3322.880127
1	GIST	2nd	IVFPQ	m=32	nprobe=64	0.2778	7186.499470
2	GIST	3rd	LSH	nbits=512	-	0.0722	1243.841426
3	GloVe	1st	HNSWSQ	SQ8	efSearch=256	0.9226	6994.501615
4	GloVe	2nd	IVFPQ	m=25	nprobe=64	0.5615	21613.287063
5	GloVe	3rd	LSH	nbits=512	-	0.4296	1066.170398
6	SIFT	1st	HNSWSQ	SQ8	efSearch=256	0.9857	14707.681881
7	SIFT	2nd	IVFPQ	m=32	nprobe=64	0.7276	13782.406213
8	SIFT	3rd	LSH	nbits=512	-	0.3790	1233.633081

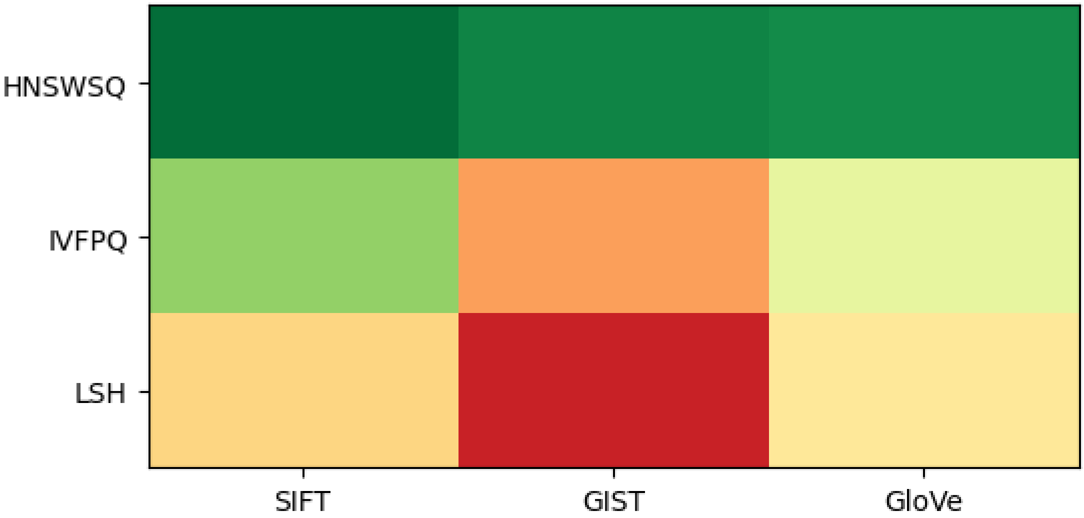
In [15]:

```
heat = (peak.pivot(index="method", columns="dataset", values="recall")
            .reindex(index=["HNSWSQ", "IVFPQ", "LSH"],
```

```
                      columns=["SIFT",   "GIST",  "GloVe"]))

fig, ax = plt.subplots(figsize=(6, 3))
im = ax.imshow(heat.values, cmap="RdYlGn", vmin=0, vmax=1, aspect="auto")
ax.set_xticks(range(len(heat.columns))); ax.set_xticklabels(heat.columns)
ax.set_yticks(range(len(heat.index)));   ax.set_yticklabels(heat.index)
```

Out[15]:

```
[Text(0, 0, 'HNSWSQ'), Text(0, 1, 'IVFPQ'), Text(0, 2, 'LSH')]
```



In [16]:

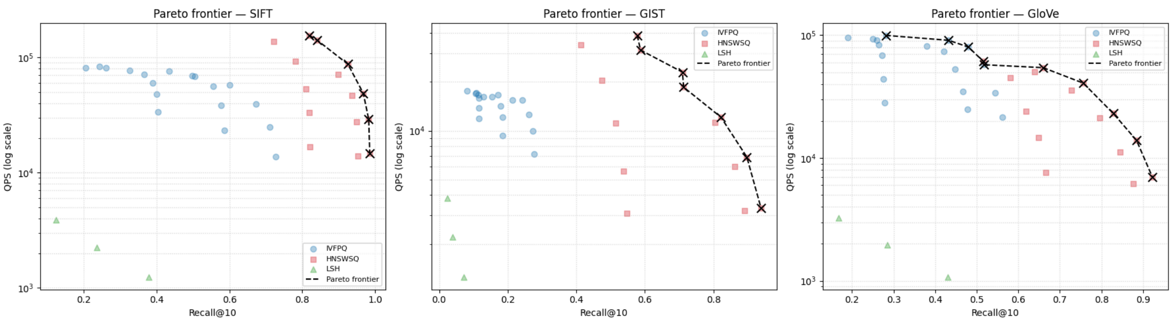
```
for i in range(heat.shape[0]):
    for j in range(heat.shape[1]):
        ax.text(j, i, f"{heat.values[i, j]:.2f}",
                ha="center", va="center", fontsize=11, fontweight="bold")

ax.set_title("Peak Recall@10 per (method × dataset)")
fig.colorbar(im, ax=ax, label="Recall@10", shrink=0.8)
fig.tight_layout()
fig.savefig(os.path.join(PLOTS_DIR, "winner_heatmap.png"), dpi=150)
plt.show()
```

5.3 Pareto frontier overlay

In [17]:

```
def pareto_mask(recalls, qpses):
    n = len(recalls)
    keep = np.ones(n, dtype=bool)
    for i in range(n):
        dominated = (((recalls >= recalls[i]) & (qpses >= qpses[i]))
                     & ((recalls >  recalls[i]) | (qpses >  qpses[i])))
        if dominated.any():
            keep[i] = False
    return keep

datasets_in_order = ["SIFT", "GIST", "GloVe"]
fig, axes = plt.subplots(1, len(datasets_in_order),
                         figsize=(6 * len(datasets_in_order), 5), sharey=False)

for ax, ds in zip(axes, datasets_in_order):
    sub = results[results.dataset == ds]

    for method in ["IVFPQ", "HNSWSQ", "LSH"]:
        d = sub[sub.method == method]
        ax.scatter(d.recall, d.qps,
```

```
                   marker=MARKERS[method], color=COLORS[method],
                   alpha=0.35, s=40, label=f"{method}")

    r = sub.recall.values
    q = sub.qps.values
    mask = pareto_mask(r, q)

    pf = sub[mask].sort_values("recall")
    ax.plot(pf.recall, pf.qps, "k--", lw=1.5, zorder=10, label="Pareto frontier"
    ax.scatter(pf.recall, pf.qps, c="black", s=90, marker="x", zorder=11)

    ax.set_yscale("log")
    ax.set_xlabel("Recall@10")
    ax.set_ylabel("QPS (log scale)")
    ax.set_title(f"Pareto frontier — {ds}")
    ax.grid(True, which="both", ls=":", alpha=0.5)
    ax.legend(fontsize=8, loc="best")

fig.tight_layout()
fig.savefig(os.path.join(PLOTS_DIR, "pareto_frontiers.png"), dpi=150)
plt.show()

frontier_rows = []
for ds in datasets_in_order:
    sub = results[results.dataset == ds]
    mask = pareto_mask(sub.recall.values, sub.qps.values)
    frontier_rows.append(sub[mask][["dataset", "method", "param1", "param2",
                                    "recall", "qps"]].sort_values("recall"))

pareto_df = pd.concat(frontier_rows, ignore_index=True)
pareto_df.to_csv(os.path.join(RESULTS_DIR, "pareto_frontier.csv"), index=False)
print("Pareto-optimal configs per dataset:",
      pareto_df.groupby("dataset").size().to_dict())
pareto_df
```




```
Pareto-optimal configs per dataset: {'GIST': 7, 'GloVe': 10, 'SIFT': 6}
```

Out[17]:

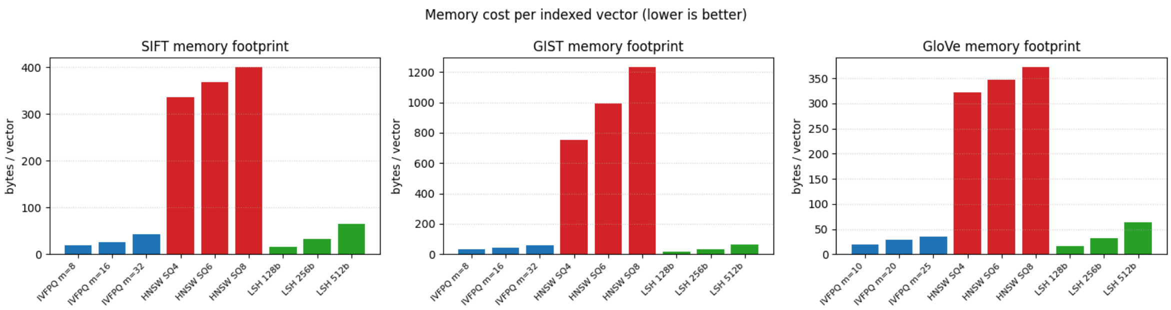
	dataset	method	param1	param2	recall	qps
0	SIFT	HNSWSQ	SQ6	efSearch=16	0.8190	155169.151776
1	SIFT	HNSWSQ	SQ8	efSearch=16	0.8411	140986.794804
2	SIFT	HNSWSQ	SQ8	efSearch=32	0.9255	87299.685251
3	SIFT	HNSWSQ	SQ8	efSearch=64	0.9679	48751.396247
4	SIFT	HNSWSQ	SQ8	efSearch=128	0.9827	28988.125928
5	SIFT	HNSWSQ	SQ8	efSearch=256	0.9857	14707.681881
6	GIST	HNSWSQ	SQ8	efSearch=16	0.5777	38746.595695
7	GIST	HNSWSQ	SQ6	efSearch=16	0.5873	31430.815806
8	GIST	HNSWSQ	SQ8	efSearch=32	0.7095	22984.870777
9	GIST	HNSWSQ	SQ6	efSearch=32	0.7130	18544.848036
10	GIST	HNSWSQ	SQ8	efSearch=64	0.8217	12116.653517
11	GIST	HNSWSQ	SQ8	efSearch=128	0.8973	6874.920477
12	GIST	HNSWSQ	SQ8	efSearch=256	0.9388	3322.880127
13	GloVe	IVFPQ	m=25	nprobe=1	0.2811	99724.948510
14	GloVe	IVFPQ	m=25	nprobe=4	0.4321	90968.195667
15	GloVe	IVFPQ	m=25	nprobe=8	0.4797	80212.299500
16	GloVe	HNSWSQ	SQ4	efSearch=16	0.5161	61177.910307
17	GloVe	IVFPQ	m=25	nprobe=16	0.5179	57509.650369
18	GloVe	HNSWSQ	SQ8	efSearch=16	0.6596	54426.614181
19	GloVe	HNSWSQ	SQ8	efSearch=32	0.7563	40729.469654
20	GloVe	HNSWSQ	SQ8	efSearch=64	0.8287	23028.642305
21	GloVe	HNSWSQ	SQ8	efSearch=128	0.8844	13933.703602
22	GloVe	HNSWSQ	SQ8	efSearch=256	0.9226	6994.501615

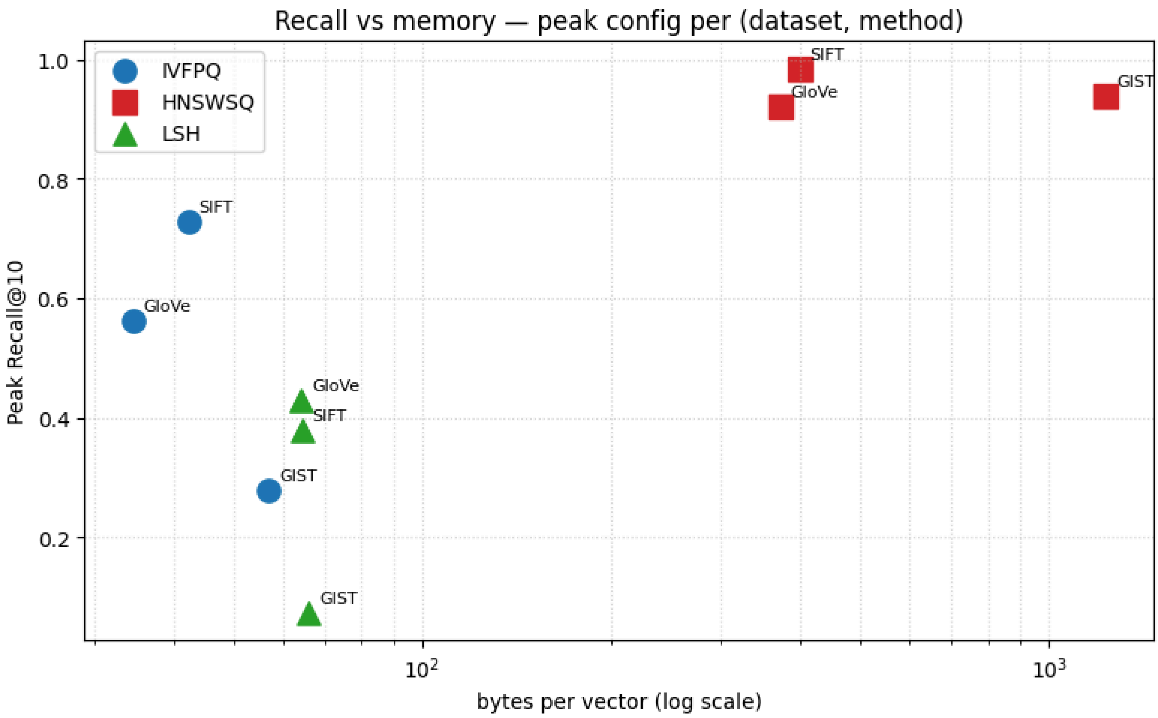
5.4 Memory footprint — the missing axis of the trade-off

In [18]:

```
mem_path = os.path.join(RESULTS_DIR, "memory_results.csv")
if not os.path.exists(mem_path):
    print(f"{mem_path} not found — run measure_memory.py on the cluster first.")
else:
    mem = pd.read_csv(mem_path)

    fig, axes = plt.subplots(1, 3, figsize=(15, 4), sharey=False)
    for ax, ds in zip(axes, ["SIFT", "GIST", "GloVe"]):
        sub = mem[mem.dataset == ds]
        labels, values, colors = [], [], []
        for _, r in sub.iterrows():
            if r.method == "IVFPQ":
                labels.append(f"IVFPQ m={int(r.m)}");  colors.append(COLORS["IVF
```

```
            elif r.method == "HNSWSQ":
                labels.append(f"HNSW SQ{int(r.sq_bits)}"); colors.append(COLORS[
            else:
                labels.append(f"LSH {int(r.nbits)}b"); colors.append(COLORS["LSH
            values.append(r.bytes_per_vector)
        ax.bar(range(len(labels)), values, color=colors)
        ax.set_xticks(range(len(labels)))
        ax.set_xticklabels(labels, rotation=45, ha="right", fontsize=8)
        ax.set_ylabel("bytes / vector")
        ax.set_title(f"{ds} memory footprint")
        ax.grid(True, axis="y", ls=":", alpha=0.5)
    fig.suptitle("Memory cost per indexed vector (lower is better)")
    fig.tight_layout()
    fig.savefig(os.path.join(PLOTS_DIR, "memory_per_vector.png"), dpi=150)
    plt.show()

    peak = (results.loc[results.groupby(["dataset", "method"])["recall"].idxmax(
            [["dataset", "method", "m", "sq_bits", "nbits", "recall"]])

    def _mem_for(row):
        q = mem[(mem.dataset == row.dataset) & (mem.method == row.method)]
        if row.method == "IVFPQ":   q = q[q.m == row.m]
        if row.method == "HNSWSQ":  q = q[q.sq_bits == row.sq_bits]
        if row.method == "LSH":     q = q[q.nbits == row.nbits]
        return q.bytes_per_vector.values[0] if len(q) else np.nan

    peak["bytes_per_vector"] = peak.apply(_mem_for, axis=1)

    fig, ax = plt.subplots(figsize=(8, 5))
    for method in ("IVFPQ", "HNSWSQ", "LSH"):
        d = peak[peak.method == method]
        ax.scatter(d.bytes_per_vector, d.recall,
                   marker=MARKERS[method], color=COLORS[method],
                   s=120, label=method)
        for _, r in d.iterrows():
            ax.annotate(r.dataset, (r.bytes_per_vector, r.recall),
                        fontsize=8, xytext=(5, 5), textcoords="offset points")
    ax.set_xscale("log")
    ax.set_xlabel("bytes per vector (log scale)")
    ax.set_ylabel("Peak Recall@10")
    ax.set_title("Recall vs memory — peak config per (dataset, method)")
    ax.grid(True, which="both", ls=":", alpha=0.5)
    ax.legend()
    fig.tight_layout()
    fig.savefig(os.path.join(PLOTS_DIR, "recall_vs_memory.png"), dpi=150)
    plt.show()
```





6. Relative Contrast measuring query hardness

Relative Contrast (RC) measures how much a query's true neighbours *stand out* from the rest of the database. For a query q, let r_k be the distance to its k-th nearest neighbour and d_{mean} the average distance from q to all database points:

$$d_{\text{mean}} = \frac{1}{|S|}\sum_{x\in S} d(q,x), \qquad \mathrm{RC}_k(q) = \frac{d_{\text{mean}}}{r_k}.$$

A **small RC** (close to 1) means the k-th neighbour is barely closer than an average point — the query is **hard**. A **large RC** means the neighbours are clearly separated — the query is **easy**.

For Euclidean datasets we use distances directly. For normalised GloVe we use the cosine-distance proxy $d(q,x) = 1 - q^\top x$, which is a proper distance on the unit sphere, so RC remains well defined.

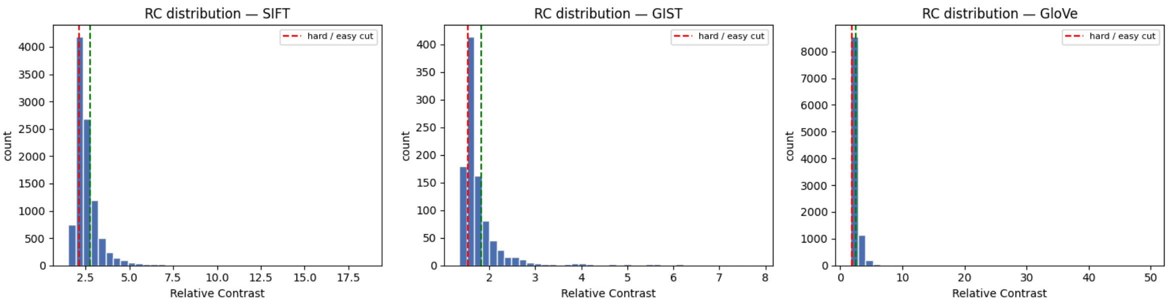
In [19]:
```
def compute_rc(xb, xq, k, metric, batch=256):
    d = xb.shape[1]
    if metric == faiss.METRIC_INNER_PRODUCT:
        gt = faiss.IndexFlatIP(d)
    else:
        gt = faiss.IndexFlatL2(d)
    gt.add(xb)
    Dk, _ = gt.search(xq, k)

    if metric == faiss.METRIC_L2:
        rk = np.sqrt(np.maximum(Dk[:, k - 1], 0.0))
    else:
        rk = 1.0 - Dk[:, k - 1]

    dmean = np.empty(xq.shape[0], dtype="float64")
    for i in range(0, xq.shape[0], batch):
```

```
        q = xq[i:i + batch]
        if metric == faiss.METRIC_L2:
            xb2 = (xb ** 2).sum(1)[None, :]
            q2  = (q ** 2).sum(1)[:, None]
            dist = np.sqrt(np.maximum(xb2 + q2 - 2.0 * q @ xb.T, 0.0))
        else:
            dist = 1.0 - q @ xb.T
        dmean[i:i + batch] = dist.mean(axis=1)

    rk = np.where(rk <= 1e-12, 1e-12, rk)
    return dmean / rk
```

In [20]:

```
RC = {}
for name in LOADED:
    RC[name] = np.load(os.path.join(QUERY_DIR, name.lower(), "rc_values.npy"))
    print(f"{name}: RC loaded  shape={RC[name].shape}  "
          f"min={RC[name].min():.3f}  median={np.median(RC[name]):.3f}  max={RC[
fig, axes = plt.subplots(1, len(RC), figsize=(5 * len(RC), 4))
if len(RC) == 1:
    axes = [axes]
for ax, (name, rc) in zip(axes, RC.items()):
    ax.hist(rc, bins=40, color="#4c72b0", edgecolor="white")
    ax.axvline(np.quantile(rc, 0.25), color="red", ls="--", label="hard / easy c
    ax.axvline(np.quantile(rc, 0.75), color="green", ls="--")
    ax.set_title(f"RC distribution — {name}")
    ax.set_xlabel("Relative Contrast"); ax.set_ylabel("count")
    ax.legend(fontsize=8)
fig.tight_layout()
fig.savefig(os.path.join(PLOTS_DIR, "rc_distributions.png"), dpi=150)
plt.show()
```

```
SIFT: RC loaded  shape=(10000,)  min=1.503  median=2.363  max=18.550
GIST: RC loaded  shape=(1000,)  min=1.352  median=1.619  max=7.837
GloVe: RC loaded  shape=(10000,)  min=1.634  median=2.096  max=49.706
```



6.1 Recall and QPS on easy vs. hard queries

We split each query set into the **bottom 25 % of RC (hard)** and the **top 25 % (easy)**, then re-measure Recall@10 and QPS for each group using a representative mid-range configuration of every method. If RC is a useful hardness predictor, the **hard** group should show **lower recall**.

In [21]:

```
REP_CONFIG = dict(
    IVFPQ  = dict(m=16, nprobe=16),
    HNSWSQ = dict(sq_bits=8, efSearch=64),
    LSH    = dict(nbits=256),
)

def split_easy_hard(rc, frac=0.25):
```

```
    """Indices of the hardest and easiest `frac` of queries."""
    n = len(rc)
    order = np.argsort(rc)
    cut = max(1, int(round(frac * n)))
    return order[:cut], order[-cut:]

def eval_on_subset(xb, xq, idx_subset, metric, method, k=K):
    """Build a representative index and evaluate recall/QPS on a query subset.""
    xq_sub = np.ascontiguousarray(xq[idx_subset])
    _, I_gt = build_ground_truth(xb, xq_sub, k, metric)
    if method == "IVFPQ":
        m = REP_CONFIG["IVFPQ"]["m"]
        m = m if xb.shape[1] % m == 0 else valid_pq_m(xb.shape[1])[0]
        index = build_ivfpq(xb, m, metric, CFG["nlist"])
        set_search_param(index, "nprobe", REP_CONFIG["IVFPQ"]["nprobe"])
    elif method == "HNSWSQ":
        index = build_hnswsq(xb, REP_CONFIG["HNSWSQ"]["sq_bits"], metric)
        set_search_param(index, "efSearch", REP_CONFIG["HNSWSQ"]["efSearch"])
    else:
        index = build_lsh(xb, REP_CONFIG["LSH"]["nbits"], metric)
    D, I, st, qps = benchmark_index(index, xq_sub, k)
    rec = recall_at_k(I, I_gt, k)
    del index; gc.collect()
    return rec, qps
```

In [22]:

```
hardness = pd.read_csv(os.path.join(RESULTS_DIR, "hardness_results.csv"))
hardness.pivot_table(index=["dataset", "method"], columns="group",
                     values="recall").round(3)
```

Out[22]:

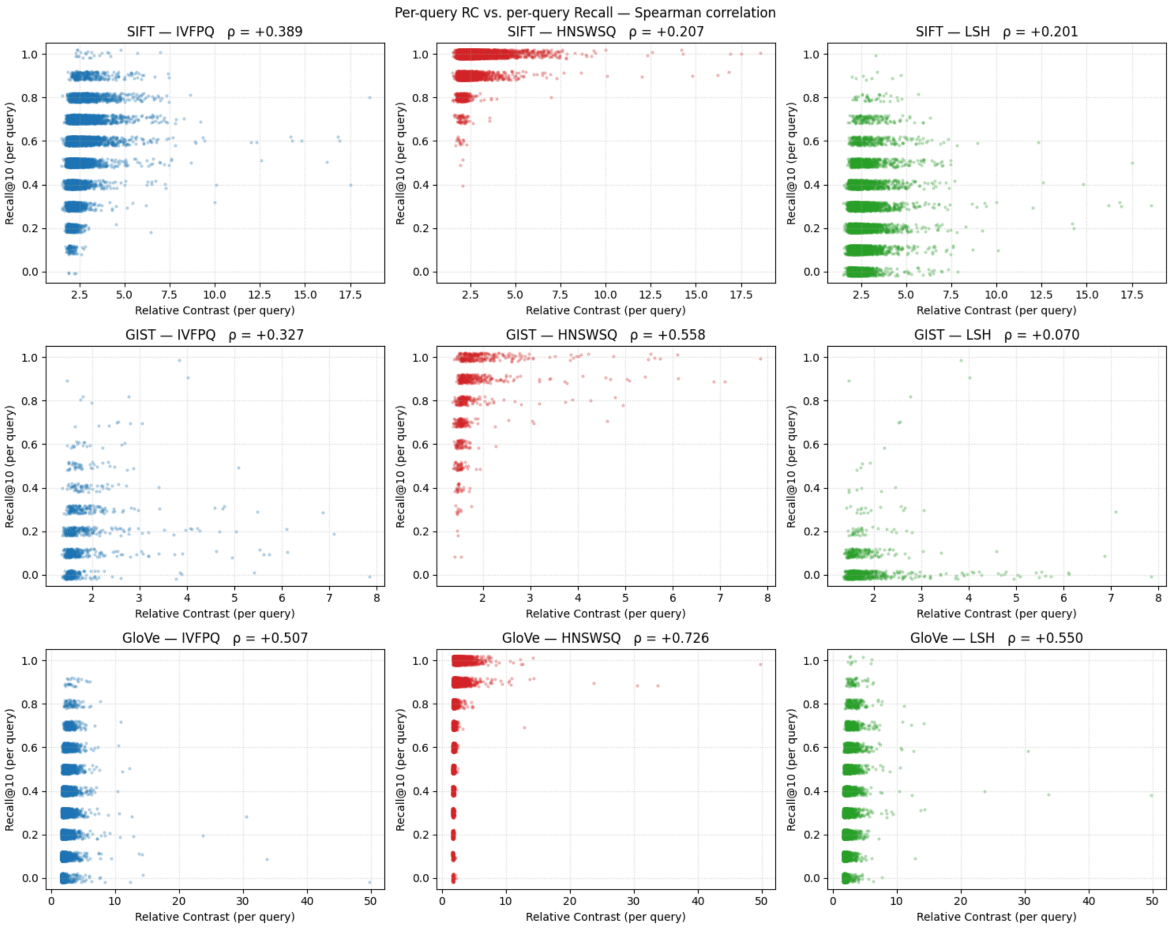
dataset	method	easy	hard
GIST	HNSWSQ	0.930	0.691
	IVFPQ	0.253	0.131
	LSH	0.059	0.031
GloVe	HNSWSQ	0.975	0.545
	IVFPQ	0.386	0.142
	LSH	0.453	0.160
SIFT	HNSWSQ	0.985	0.951
	IVFPQ	0.648	0.476
	LSH	0.295	0.195

6.2 RC ↔ recall correlation

In [23]:

```
from scipy.stats import spearmanr

corr_rows = []
fig, axes = plt.subplots(len(LOADED), 3, figsize=(15, 4 * len(LOADED)),
                         squeeze=False)
for row, ds in enumerate(LOADED):
    rc_path = os.path.join(QUERY_DIR, ds.lower(), "rc_values.npy")
```

```
    if not os.path.exists(rc_path):
        print(f"{ds}: rc_values.npy missing — run run_analysis.py first.")
        continue
    rc_vals = np.load(rc_path)

    for col, method in enumerate(("IVFPQ", "HNSWSQ", "LSH")):
        ax = axes[row][col]
        pqr_path = os.path.join(QUERY_DIR, ds.lower(),
                                f"per_query_recall_{method}.npy")
        if not os.path.exists(pqr_path):
            ax.set_visible(False); continue
        pqr = np.load(pqr_path)

        rho, pval = spearmanr(rc_vals, pqr)
        corr_rows.append(dict(dataset=ds, method=method,
                              spearman_rho=rho, p_value=pval))

        jitter = (np.random.RandomState(0).rand(len(pqr)) - 0.5) * 0.04
        ax.scatter(rc_vals, pqr + jitter, s=4, alpha=0.25,
                   color=COLORS[method])
        ax.set_xlabel("Relative Contrast (per query)")
        ax.set_ylabel("Recall@10 (per query)")
        ax.set_title(f"{ds} — {method}   ρ = {rho:+.3f}")
        ax.set_ylim(-0.05, 1.05)
        ax.grid(True, ls=":", alpha=0.5)

fig.suptitle("Per-query RC vs. per-query Recall — Spearman correlation")
fig.tight_layout()
fig.savefig(os.path.join(PLOTS_DIR, "rc_vs_recall_correlation.png"), dpi=150)
plt.show()

corr_df = pd.DataFrame(corr_rows)
corr_df.to_csv(os.path.join(RESULTS_DIR, "rc_correlation.csv"), index=False)
print("\nSpearman ρ summary (higher = RC predicts recall better):")
print(corr_df.pivot(index="dataset", columns="method",
                    values="spearman_rho").round(3).to_string())
```




```
Spearman ρ summary (higher = RC predicts recall better):
method   HNSWSQ  IVFPQ    LSH
dataset
GIST      0.558  0.327  0.070
GloVe     0.726  0.507  0.550
SIFT      0.207  0.389  0.201
```

7. Making queries harder with Gaussian perturbation

To probe robustness we make the queries deliberately harder and watch how each method degrades. We use the simplest of the strategies suggested in the brief: **add isotropic Gaussian noise** to every query vector, $\tilde{q} = q + \sigma\,\varepsilon,\ \varepsilon \sim \mathcal{N}(0, I)$, sweeping the noise level σ.

Two important details:

- The noise scale is **relative to the data** — we express σ as a fraction of the mean query-vector norm, so the same σ is comparable across SIFT, GIST and GloVe.
- For GloVe (angular) the perturbed query is **re-normalised** so it stays on the unit sphere, otherwise inner product would no longer equal cosine similarity.

For each noise level the **ground truth is recomputed on the perturbed queries** — we are asking "can the index still find the *true* neighbours of the *noisy* query", which is the meaningful robustness question.

In [24]:
```
def add_noise(xq, sigma_rel, metric):
    mean_norm = float(np.linalg.norm(xq, axis=1).mean())
    sigma = sigma_rel * mean_norm
    noisy = xq + sigma * np.random.randn(*xq.shape).astype("float32")
    noisy = noisy.astype("float32")
    if metric == faiss.METRIC_INNER_PRODUCT:
        noisy = l2_normalize(noisy)
    return np.ascontiguousarray(noisy)

NOISE_LEVELS = [0.00, 0.01, 0.05, 0.10]
```

In [25]:
```
def perturbation_study(loaded, k=K):
    rows = []
    for name, data in loaded.items():
        xb, xq, metric = data["xb"], data["xq"], data["metric"]
        print(f"=== {name} ===")
        # build representative indexes once; reuse across noise levels
        m = REP_CONFIG["IVFPQ"]["m"]
        m = m if xb.shape[1] % m == 0 else valid_pq_m(xb.shape[1])[0]
        idx_ivf = build_ivfpq(xb, m, metric, CFG["nlist"])
        set_search_param(idx_ivf, "nprobe", REP_CONFIG["IVFPQ"]["nprobe"])
        idx_hnsw = build_hnswsq(xb, REP_CONFIG["HNSWSQ"]["sq_bits"], metric)
        set_search_param(idx_hnsw, "efSearch", REP_CONFIG["HNSWSQ"]["efSearch"])
        idx_lsh = build_lsh(xb, REP_CONFIG["LSH"]["nbits"], metric)
        indexes = {"IVFPQ": idx_ivf, "HNSWSQ": idx_hnsw, "LSH": idx_lsh}

        for sigma in NOISE_LEVELS:
            xq_n = xq if sigma == 0.0 else add_noise(xq, sigma, metric)
            _, I_gt = build_ground_truth(xb, xq_n, k, metric)
            mean_rc = float(compute_rc(xb, xq_n, k, metric).mean())
            for method, idx in indexes.items():
                D, I, st, qps = benchmark_index(idx, xq_n, k)
                rows.append(dict(dataset=name, method=method, sigma=sigma,
                                 recall=recall_at_k(I, I_gt, k), qps=qps,
                                 mean_rc=mean_rc))
            print(f"  sigma={sigma:.2f}  mean_RC={mean_rc:.3f}")
        for idx in indexes.values():
            del idx
        gc.collect()
    return pd.DataFrame(rows)

perturb = pd.read_csv(os.path.join(RESULTS_DIR, "perturbation_results.csv"))
if "level" not in perturb.columns and "sigma" in perturb.columns:
    perturb = perturb.rename(columns={"sigma": "level"})
    perturb["strategy"] = "gaussian"

perturb.pivot_table(index=["dataset", "strategy", "level"], columns="method",
                    values="recall").round(3)
```

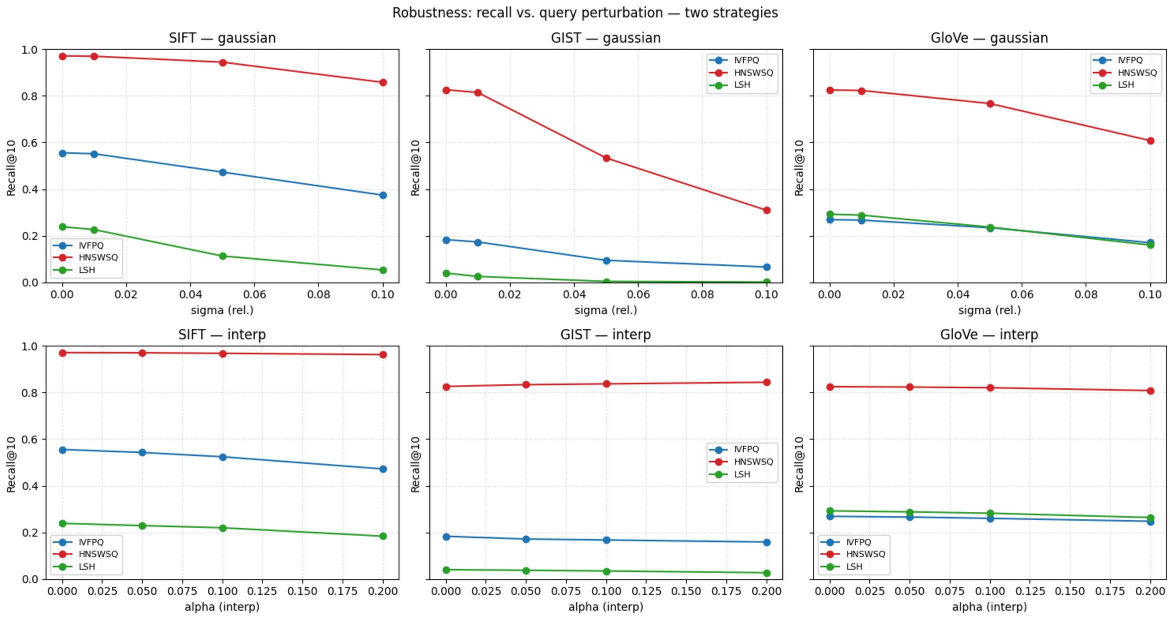
Out[25]:

		method	HNSWSQ	IVFPQ	LSH
dataset	**strategy**	**level**			
GIST	**gaussian**	**0.00**	0.826	0.183	0.040
		0.01	0.815	0.173	0.025
		0.05	0.534	0.094	0.004
		0.10	0.310	0.066	0.001
	interp	**0.00**	0.826	0.183	0.040
		0.05	0.834	0.171	0.037
		0.10	0.837	0.167	0.034
		0.20	0.844	0.158	0.027
GloVe	**gaussian**	**0.00**	0.825	0.269	0.292
		0.01	0.823	0.267	0.288
		0.05	0.767	0.234	0.237
		0.10	0.608	0.170	0.160
	interp	**0.00**	0.825	0.269	0.292
		0.05	0.823	0.266	0.288
		0.10	0.821	0.260	0.282
		0.20	0.809	0.248	0.264
SIFT	**gaussian**	**0.00**	0.971	0.556	0.239
		0.01	0.970	0.552	0.226
		0.05	0.945	0.473	0.113
		0.10	0.858	0.374	0.053
	interp	**0.00**	0.971	0.556	0.239
		0.05	0.971	0.543	0.229
		0.10	0.968	0.524	0.220
		0.20	0.963	0.472	0.184

In [26]:

```
strategies = [s for s in ("gaussian", "interp") if s in perturb.strategy.unique(
fig, axes = plt.subplots(len(strategies), len(LOADED),
                         figsize=(5 * len(LOADED), 4 * len(strategies)),
                         sharey=True, squeeze=False)
for row, strategy in enumerate(strategies):
    for col, name in enumerate(LOADED):
        ax = axes[row][col]
        sub = perturb[(perturb.dataset == name) & (perturb.strategy == strategy)
        for method in ("IVFPQ", "HNSWSQ", "LSH"):
            d = sub[sub.method == method].sort_values("level")
            ax.plot(d.level, d.recall, marker="o", color=COLORS[method],
                    label=method)
```

```
        ax.set_title(f"{name} — {strategy}")
        ax.set_xlabel("sigma (rel.)" if strategy == "gaussian" else "alpha (inte
        ax.set_ylabel("Recall@10"); ax.set_ylim(0, 1)
        ax.grid(True, ls=":", alpha=0.5); ax.legend(fontsize=8)
fig.suptitle("Robustness: recall vs. query perturbation — two strategies")
fig.tight_layout()
fig.savefig(os.path.join(PLOTS_DIR, "perturbation_recall.png"), dpi=150)
plt.show()
```



8. Can we adapt to hard queries?

We test this directly: take the hardest 25 % of queries (lowest RC) and re-run them while **increasing the search-time knob** of the two tunable methods (`nprobe` for IVFPQ, `efSearch` for HNSWSQ). LSH has no search-time knob, so it is shown only as a baseline.

In [27]:
```
def adaptation_study(loaded, k=K):
    rows = []
    for name, data in loaded.items():
        xb, xq, metric = data["xb"], data["xq"], data["metric"]
        hard_idx, _ = split_easy_hard(RC[name])
        xq_hard = np.ascontiguousarray(xq[hard_idx])
        _, I_gt = build_ground_truth(xb, xq_hard, k, metric)

        m = REP_CONFIG["IVFPQ"]["m"]
        m = m if xb.shape[1] % m == 0 else valid_pq_m(xb.shape[1])[0]
        idx = build_ivfpq(xb, m, metric, CFG["nlist"])
        for nprobe in NPROBE_VALUES:
            set_search_param(idx, "nprobe", nprobe)
            D, I, st, qps = benchmark_index(idx, xq_hard, k)
            rows.append(dict(dataset=name, method="IVFPQ", knob="nprobe",
                             value=nprobe, recall=recall_at_k(I, I_gt, k),
                             qps=qps))
        del idx; gc.collect()

        idx = build_hnswsq(xb, REP_CONFIG["HNSWSQ"]["sq_bits"], metric)
        for ef in EFSEARCH_VALUES:
            set_search_param(idx, "efSearch", ef)
```

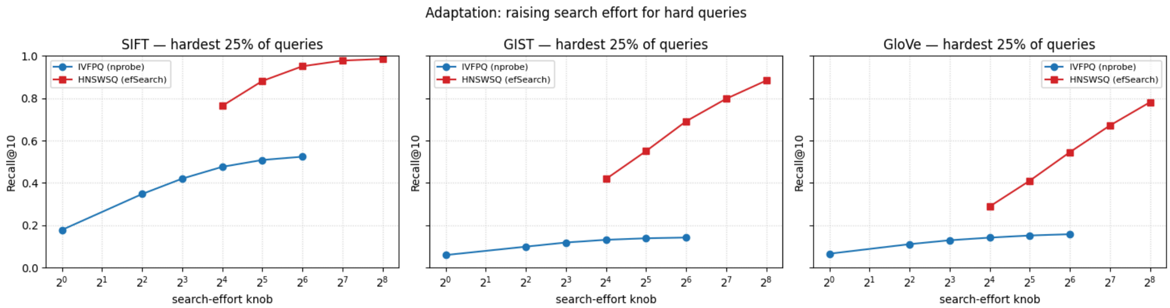
```
            D, I, st, qps = benchmark_index(idx, xq_hard, k)
            rows.append(dict(dataset=name, method="HNSWSQ", knob="efSearch",
                             value=ef, recall=recall_at_k(I, I_gt, k),
                             qps=qps))
        del idx; gc.collect()
        print(f"{name}: adaptation sweep done")
    return pd.DataFrame(rows)

adapt = pd.read_csv(os.path.join(RESULTS_DIR, "adaptation_results.csv"))
adapt.head(12)
```

Out[27]:

	dataset	method	knob	value	recall	qps
0	SIFT	IVFPQ	nprobe	1	0.17796	154182.661062
1	SIFT	IVFPQ	nprobe	4	0.34736	147770.608402
2	SIFT	IVFPQ	nprobe	8	0.42012	127001.634812
3	SIFT	IVFPQ	nprobe	16	0.47584	103940.038353
4	SIFT	IVFPQ	nprobe	32	0.50832	71328.739726
5	SIFT	IVFPQ	nprobe	64	0.52384	42869.038891
6	SIFT	HNSWSQ	efSearch	16	0.76368	165302.567727
7	SIFT	HNSWSQ	efSearch	32	0.88140	112698.466489
8	SIFT	HNSWSQ	efSearch	64	0.95096	60082.414339
9	SIFT	HNSWSQ	efSearch	128	0.97784	32832.648143
10	SIFT	HNSWSQ	efSearch	256	0.98560	15992.255859
11	GIST	IVFPQ	nprobe	1	0.05920	31789.960422

In [28]:

```
fig, axes = plt.subplots(1, len(LOADED), figsize=(5 * len(LOADED), 4), sharey=Tr
if len(LOADED) == 1:
    axes = [axes]
for ax, name in zip(axes, LOADED):
    sub = adapt[adapt.dataset == name]
    for method, mark in (("IVFPQ", "o"), ("HNSWSQ", "s")):
        d = sub[sub.method == method].sort_values("value")
        ax.plot(d.value, d.recall, marker=mark, color=COLORS[method],
                label=f"{method} ({d.knob.iloc[0]})")
    ax.set_xscale("log", base=2)
    ax.set_title(f"{name} — hardest 25% of queries")
    ax.set_xlabel("search-effort knob"); ax.set_ylabel("Recall@10")
    ax.set_ylim(0, 1); ax.grid(True, which="both", ls=":", alpha=0.5)
    ax.legend(fontsize=8)
fig.suptitle("Adaptation: raising search effort for hard queries")
fig.tight_layout()
fig.savefig(os.path.join(PLOTS_DIR, "adaptation_hard_queries.png"), dpi=150)
plt.show()
```



9. Exporting the modified query sets

In [29]:
```python
def write_fvecs(path, arr):
    """Write a float32 array to .fvecs format."""
    arr = np.ascontiguousarray(arr.astype("float32"))
    n, d = arr.shape
    out = np.empty((n, d + 1), dtype="float32")
    out[:, 0] = np.frombuffer(np.int32(d).tobytes(), dtype="float32")[0]
    out[:, 1:] = arr
    out.tofile(path)

for name, data in LOADED.items():
    xq, metric = data["xq"], data["metric"]
    folder = os.path.join(QUERY_DIR, name.lower())
    os.makedirs(folder, exist_ok=True)
    for sigma in NOISE_LEVELS:
        xq_n = xq if sigma == 0.0 else add_noise(xq, sigma, metric)
        write_fvecs(os.path.join(folder, f"query_sigma{sigma:.2f}.fvecs"), xq_n)
    hard_idx, easy_idx = split_easy_hard(RC[name])
    np.save(os.path.join(folder, "hard_idx.npy"), hard_idx)
    np.save(os.path.join(folder, "easy_idx.npy"), easy_idx)
    np.save(os.path.join(folder, "rc_values.npy"), RC[name])
    print(f"{name}: modified queries written to {folder}")
```

```
SIFT: modified queries written to ./modified_queries\sift
GIST: modified queries written to ./modified_queries\gist
GloVe: modified queries written to ./modified_queries\glove
```

10. Summary of findings

In [30]:
```python
print("BEST CONFIGURATION PER METHOD (highest QPS at Recall@10 >= 0.90)\n")
print(best.to_string(index=False))

print("\n\nEASY vs HARD RECALL GAP (representative configs)\n")
gap = hardness.pivot_table(index=["dataset", "method"], columns="group",
                           values="recall")
gap["gap"] = (gap["easy"] - gap["hard"]).round(3)
print(gap.round(3).to_string())

print("\n\nRECALL AT HIGHEST PERTURBATION vs CLEAN (per strategy)\n")
for strat in sorted(perturb.strategy.unique()):
    sub = perturb[perturb.strategy == strat]
    piv = sub.pivot_table(index=["dataset", "method"], columns="level",
                          values="recall")
    print(f"\n--- strategy = {strat} ---")
    print(piv.round(3).to_string())
```

```
BEST CONFIGURATION PER METHOD (highest QPS at Recall@10 >= 0.90)

dataset method    param1       param2  recall          qps  met_target
   GIST HNSWSQ       SQ8 efSearch=256  0.9388  3322.880127        True
   GIST  IVFPQ      m=32    nprobe=64  0.2778  7186.499470       False
   GIST    LSH nbits=512            -  0.0722  1243.841426       False
  GloVe HNSWSQ       SQ8 efSearch=256  0.9226  6994.501615        True
  GloVe  IVFPQ      m=25    nprobe=64  0.5615 21613.287063       False
  GloVe    LSH nbits=512            -  0.4296  1066.170398       False
   SIFT HNSWSQ       SQ8  efSearch=32  0.9255 87299.685251        True
   SIFT  IVFPQ      m=32    nprobe=64  0.7276 13782.406213       False
   SIFT    LSH nbits=512            -  0.3790  1233.633081       False


EASY vs HARD RECALL GAP (representative configs)

group            easy   hard    gap
dataset method
GIST    HNSWSQ  0.930  0.691  0.238
        IVFPQ   0.253  0.131  0.122
        LSH     0.059  0.031  0.028
GloVe   HNSWSQ  0.975  0.545  0.429
        IVFPQ   0.386  0.142  0.244
        LSH     0.453  0.160  0.294
SIFT    HNSWSQ  0.985  0.951  0.034
        IVFPQ   0.648  0.476  0.172
        LSH     0.295  0.195  0.100


RECALL AT HIGHEST PERTURBATION vs CLEAN (per strategy)


--- strategy = gaussian ---
level            0.00   0.01   0.05   0.10
dataset method
GIST    HNSWSQ  0.826  0.815  0.534  0.310
        IVFPQ   0.183  0.173  0.094  0.066
        LSH     0.040  0.025  0.004  0.001
GloVe   HNSWSQ  0.825  0.823  0.767  0.608
        IVFPQ   0.269  0.267  0.234  0.170
        LSH     0.292  0.288  0.237  0.160
SIFT    HNSWSQ  0.971  0.970  0.945  0.858
        IVFPQ   0.556  0.552  0.473  0.374
        LSH     0.239  0.226  0.113  0.053

--- strategy = interp ---
level            0.00   0.05   0.10   0.20
dataset method
GIST    HNSWSQ  0.826  0.834  0.837  0.844
        IVFPQ   0.183  0.171  0.167  0.158
        LSH     0.040  0.037  0.034  0.027
GloVe   HNSWSQ  0.825  0.823  0.821  0.809
        IVFPQ   0.269  0.266  0.260  0.248
        LSH     0.292  0.288  0.282  0.264
SIFT    HNSWSQ  0.971  0.971  0.968  0.963
        IVFPQ   0.556  0.543  0.524  0.472
        LSH     0.239  0.229  0.220  0.184
```